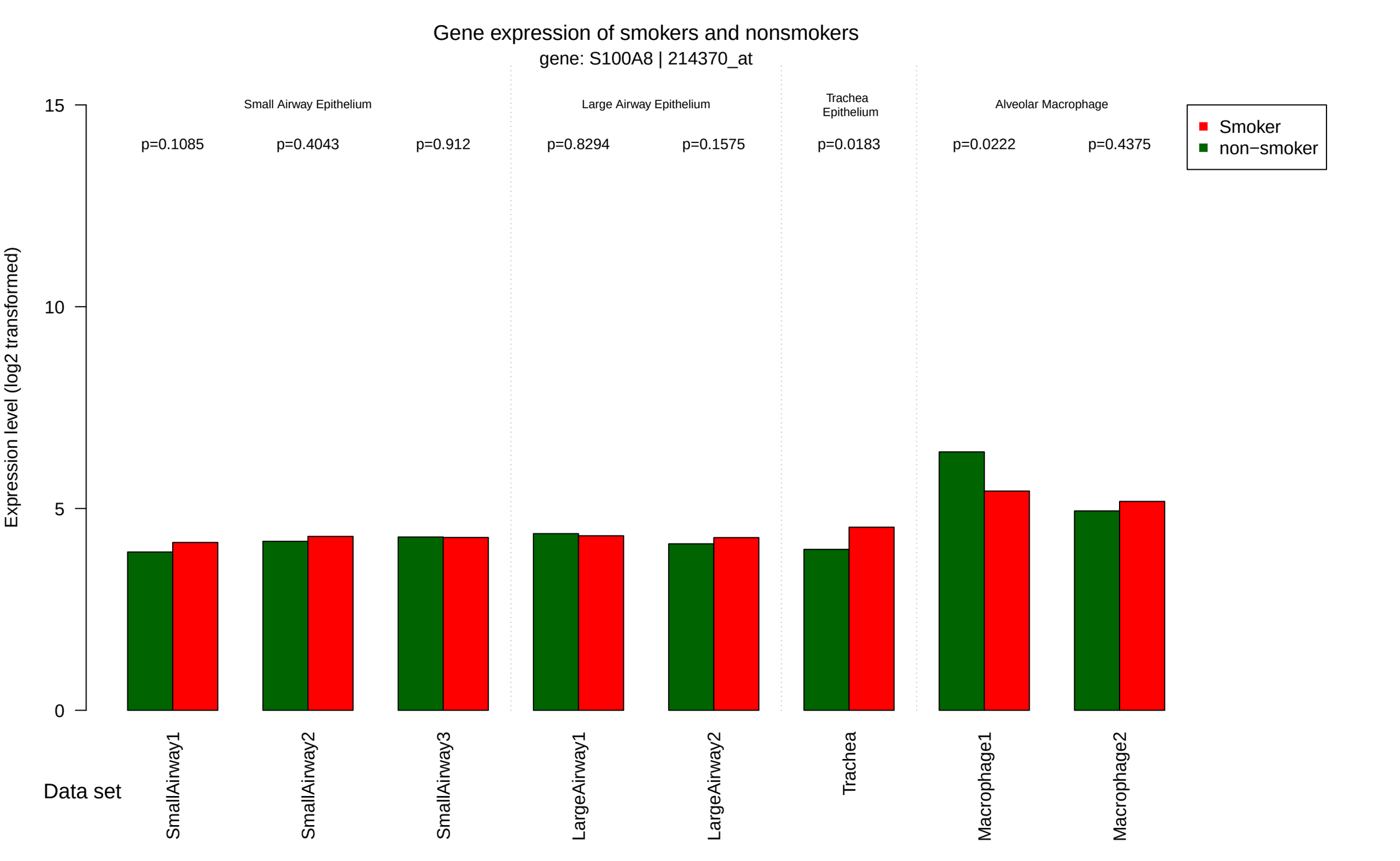
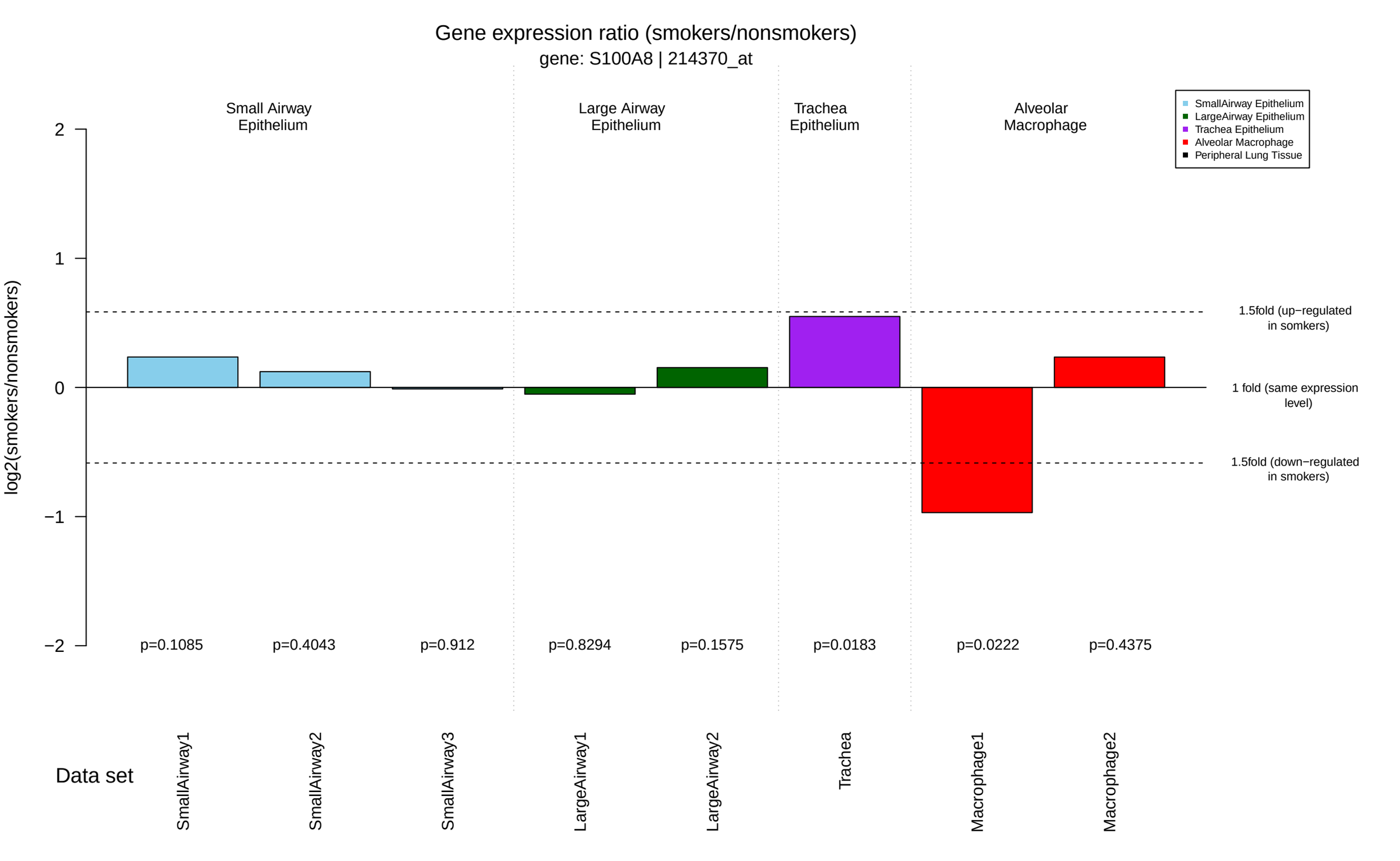
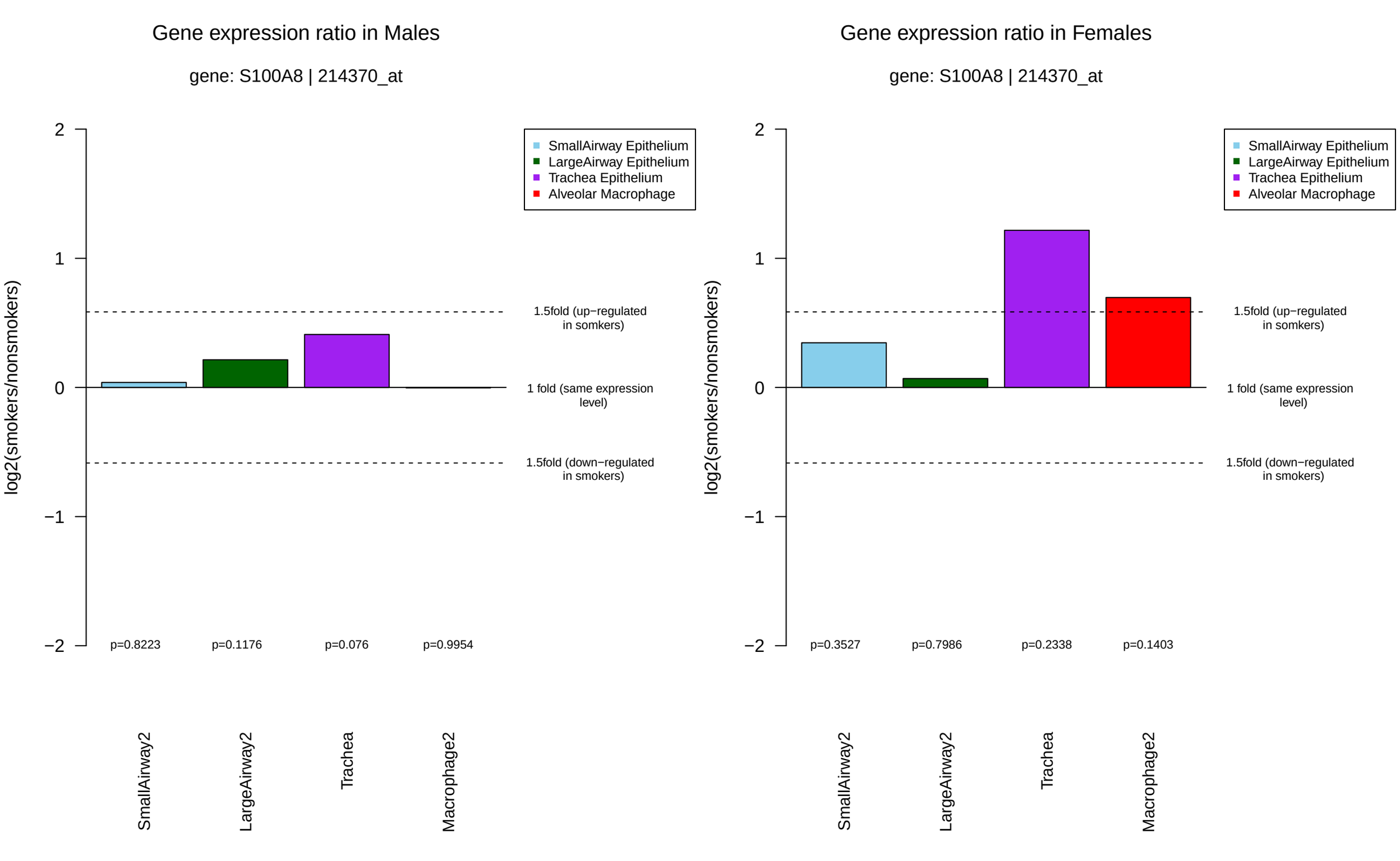
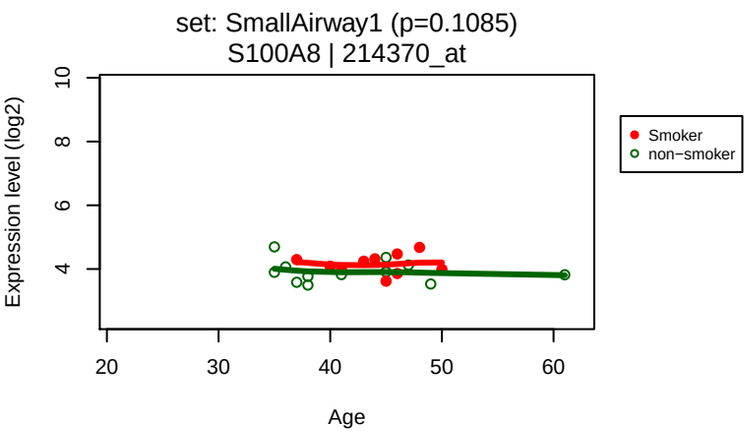
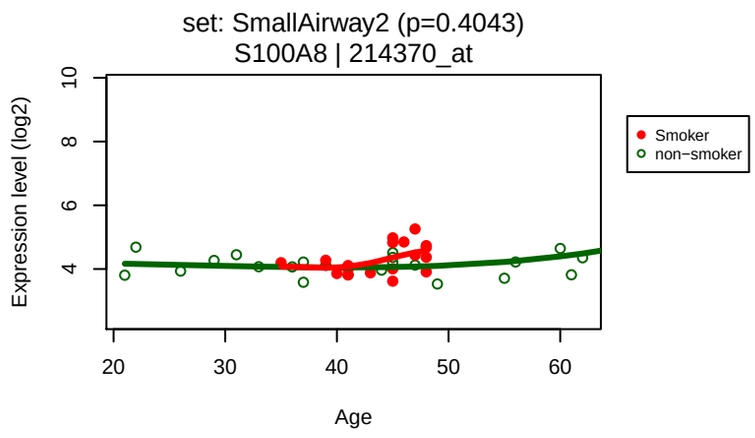
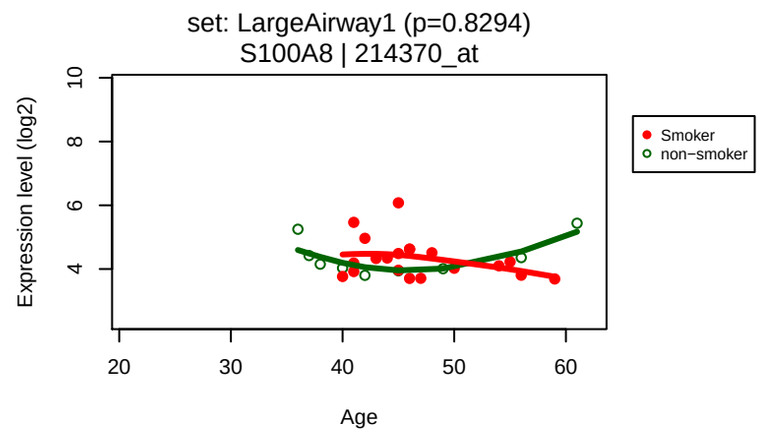

Gene expression of smokers and nonsmokers
gene: S100A8 | 214370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1085
p=0.4043
p=0.912
p=0.8294
p=0.1575
p=0.0183
p=0.0222
p=0.4375
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

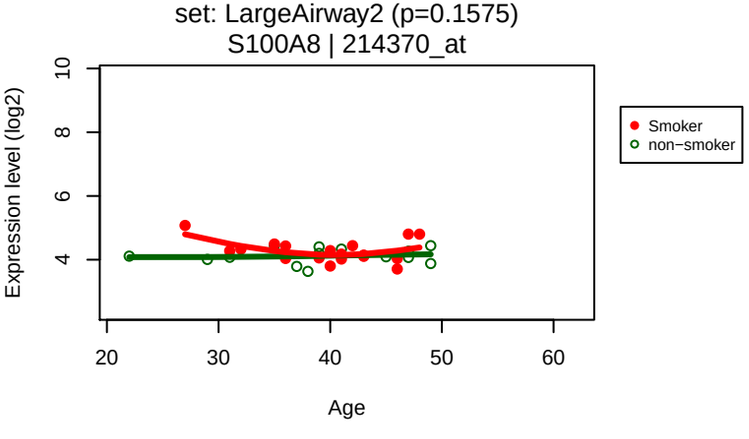

Gene expression ratio (smokers/nonsmokers)
gene: S100A8 | 214370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1085
p=0.4043
p=0.912
p=0.8294
p=0.1575
p=0.0183
p=0.0222
p=0.4375
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

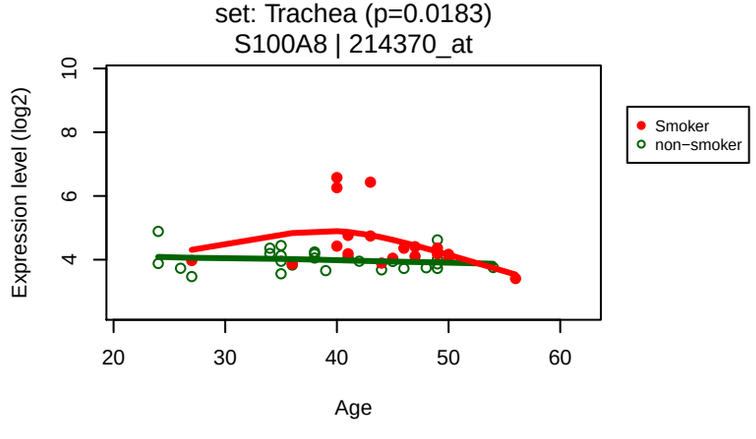

Gene expression ratio in Males
gene: S100A8 | 214370_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8223
p=0.1176
p=0.076
p=0.9954
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: S100A8 | 214370_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3527
p=0.7986
p=0.2338
p=0.1403
SmallAirway2
LargeAirway2
Trachea
Macrophage2

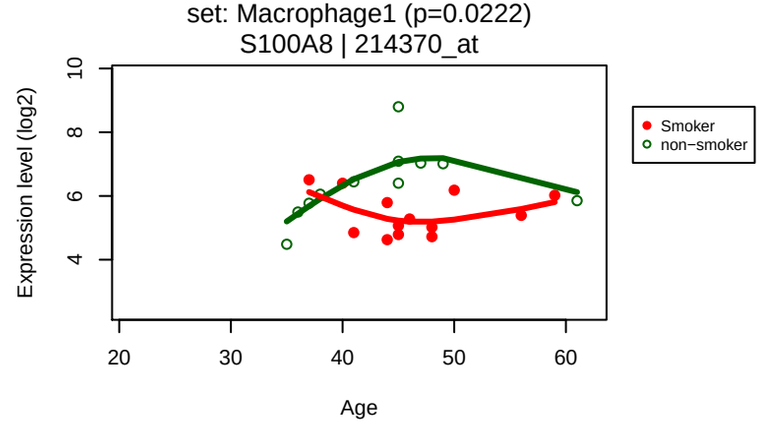

set: SmallAirway1 (p=0.1085)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker

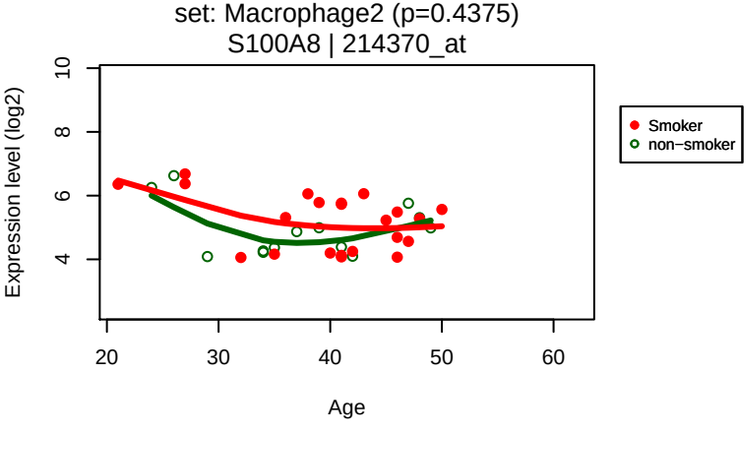

set: SmallAirway2 (p=0.4043)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker

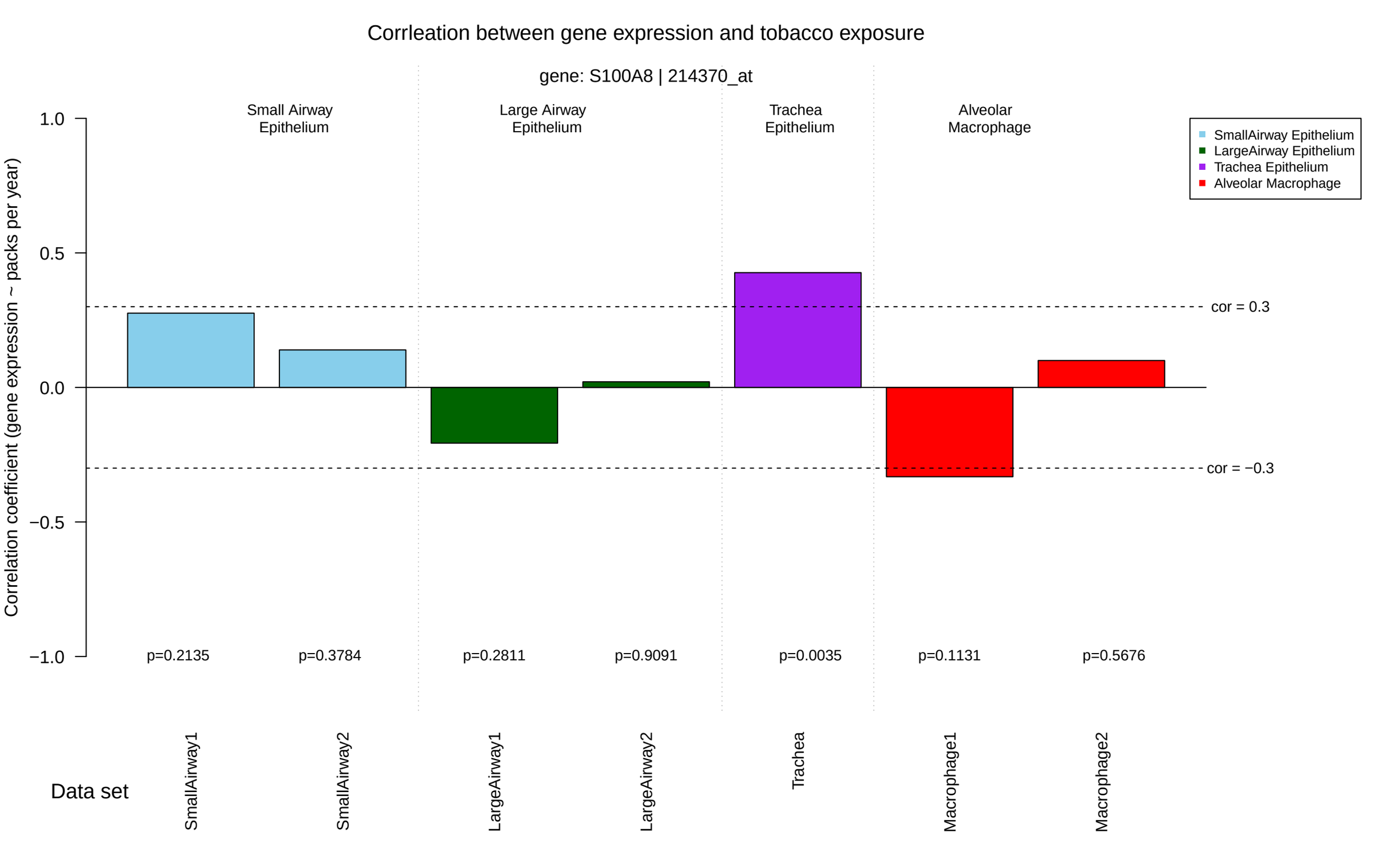

set: LargeAirway1 (p=0.8294)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1575)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0183)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0222)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4375)
S100A8 | 214370_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: S100A8 | 214370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2135
p=0.3784
p=0.2811
p=0.9091
p=0.0035
p=0.1131
p=0.5676
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2